Big Data Analysis Project Report

Beaver Group

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Introduction

The primary objective of this report is to present the insights gained from applying Big Data Analysis and Machine Learning techniques to three distinct datasets: bank credit records, wine quality evaluations, and crime reports from Los Angeles. Each team member contributed equally, with a specific area of focus: Tymofii concentrated on analyzing the bank credit data, David identified and examined key features within the LA crime reports, and Artem investigated how various factors influence wine quality.

Data Source and Description

The analysis utilized three distinct datasets: banking, wine quality, and crime data.

The banking dataset captures client interactions, including demographics, account activity, and loan statuses, offering insights into customer behavior and financial trends. The wine quality datasets, covering red and white wine, include chemical attributes like acidity, pH, and alcohol content, alongside quality scores, enabling an exploration of factors influencing wine production and consumer preferences. The crime dataset records incidents in Los Angeles, detailing crime type, location, date, and time, providing a basis for analyzing urban safety trends and geographic patterns. Each dataset supports targeted insights into its domain.

Tools

To achieve these results, we utilized a robust data analysis pipeline within the Databricks platform, which enabled scalable data processing and collaborative exploration. Leveraging PySpark, a Python API for Apache Spark, we efficiently handled large datasets, applied distributed computing, and implemented machine learning models such as linear regression and random forest to derive insights. Jupyter Notebooks facilitated an interactive workflow for data manipulation, visualization, and exploratory analysis, enhancing our understanding of the data. This combination of tools ensured accuracy, scalability, and reproducibility throughout the project.

1. Data Crime Analysis

Our objective is to analyze patterns and trends in crime data, focusing on temporal, spatial, and demographic aspects, with the goal of identifying actionable insights and recommendations.

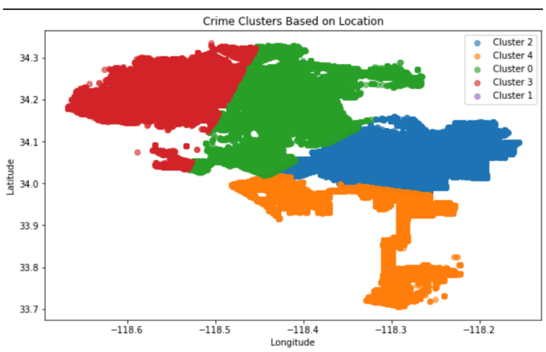
The analysis of the crime dataset reveals significant patterns and insights across temporal, demographic, and locational dimensions. Temporal factors such as the time of year, day of the week, and hour of the day demonstrate clear variations in crime frequency. For instance, Fridays and Saturdays consistently record the highest crime rates, particularly during the evening hours. Monthly trends highlight a decline in crime rates from November to December, suggesting seasonal influences.

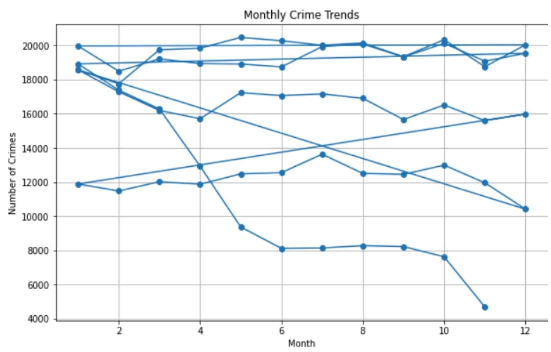
Geographic analysis uncovers distinct crime hotspots through clustering, emphasizing areas with high criminal density that could benefit from targeted law enforcement strategies. Streets and public spaces are the most common crime locations, often involving physical force ("strong-arm" tactics) or unknown weapons. These findings underline the importance of location-aware crime prevention initiatives.

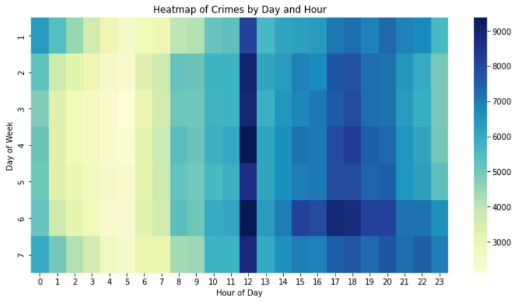
Demographically, adults aged 30-60 are the most frequent victims, and a higher proportion of crimes are reported against male victims. Ethnic distribution among victims appears balanced, with no clear overrepresentation across groups. These insights suggest opportunities to design demographic-specific interventions to reduce victimization risks.

The visualizations, including a line chart for monthly trends, a heatmap for crime activity by day and hour, Crime clusters based on location, provide a comprehensive view of the patterns. These figures highlight the temporal and spatial dynamics that should inform future strategies.

In conclusion, the findings suggest that proactive measures such as enhanced weekend patrols, focused community outreach in high-crime neighborhoods, and the use of predictive analytics could significantly improve crime prevention efforts. Tailoring interventions to temporal and locational trends ensures a more effective allocation of resources and better outcomes for public safety.







2. Bank Campaign Success Analysis

The analysis reveals significant variations in campaign success rates depending on various factors. Temporal parameters, such as the month and day of contact, play a notable role, with specific combinations showing higher success rates. Additionally, call duration proves to be a critical determinant, as successful campaigns have significantly longer average call durations compared to unsuccessful ones. Previous interactions also influence outcomes; clients with a history of positive engagements are more likely to respond favorably to the campaign and tend to have higher average account balances.

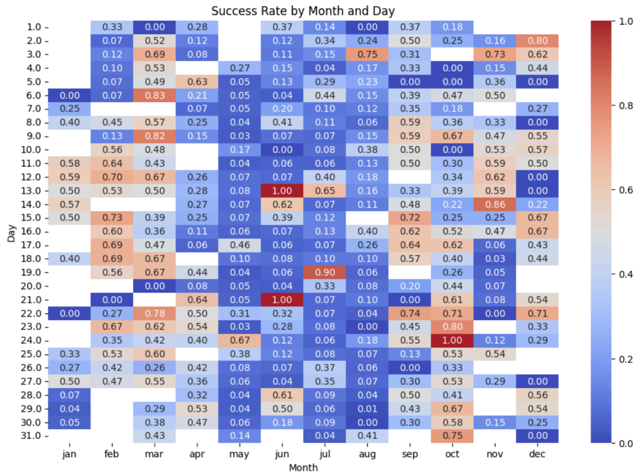


Figure 2.1 - Success rate by Month and Day of Campaign

Socio-demographic factors further highlight key differences. Clients with higher education levels (tertiary education) exhibit a higher probability of success compared to those with lower levels of education. Similarly, financial indicators like housing and loan status have a strong impact on client receptiveness. Clients without housing loans or other financial liabilities demonstrate higher success rates. These findings suggest that financial stability plays a role in the decision-making process. The timing and context of outreach, therefore, should take these aspects into consideration to optimize campaign outcomes.

Moreover, the type of profession and marital status are also influential. Students and retirees show the highest likelihood of engaging positively, while professionals in blue-collar and service industries tend to have lower success rates. Among marital statuses, single and divorced clients are more receptive compared to married individuals. These insights underline the importance of tailoring the approach to different client segments based on their socio-economic and professional profiles, alongside leveraging data-driven timing strategies to maximize engagement and conversion rates.

3. Wine Quality Analysis

The analysis of the combined wine dataset reveals key differences in chemical properties and their impact on wine quality for red and white wines. Alcohol content emerges as a significant factor, with higher quality wines generally showing higher average alcohol levels across both red and white varieties. Meanwhile, pH levels exhibit minimal variation with quality but remain consistently higher for red wines than white wines. Residual sugar levels are particularly distinctive, with white wines displaying significantly higher levels than red wines, especially in medium-quality ranges (e.g., quality level 5), reflecting differing production processes or styles.

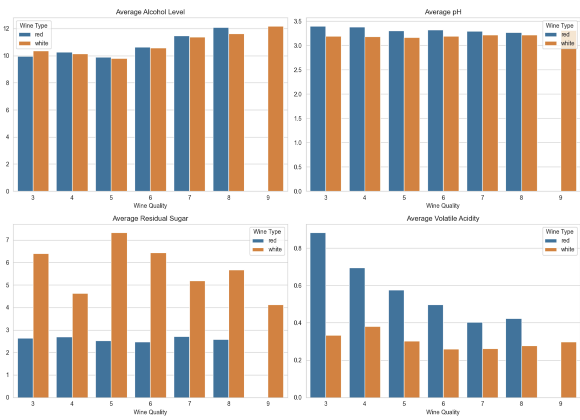


Figure 3.1 - Graphs of Level of each Parameter by Wine Quality

Volatile acidity also plays a critical role, particularly for red wines, where higher quality correlates with lower volatile acidity, unlike white wines, which show less variation in this attribute. The overall Root Mean Square Error (RMSE) of 0.7347 for the combined dataset indicates a reasonably strong predictive ability for modeling wine quality based on these properties. These findings provide valuable insights for wine producers, highlighting the need to optimize alcohol levels, residual sugar, and acidity differently for red and white wines to enhance quality and meet consumer expectations.

Conclusion

In summary, the analyses highlight actionable insights across different domains. Crime analysis emphasizes targeted interventions based on temporal and spatial trends, such as increased patrols during weekends and in high-crime areas. Bank campaign success depends on personalized outreach, leveraging factors like call timing, client demographics, and financial stability. Wine quality is influenced by key chemical properties, with alcohol content, residual sugar, and acidity playing critical roles. These findings showcase the importance of data-driven strategies to optimize outcomes effectively.

References

1. https://archive.ics.uci.edu/dataset/186/wine+quality
2. https://catalog.data.gov/dataset/crime-data-from-2020-to-present
3. https://archive.ics.uci.edu/dataset/222/bank+marketing

Annex I

Columns and their data types of:

bank-full.csv: age: integer, job: string, marital: string, education: string, default: string, balance: integer, housing: string, loan: string, contact: string, day: integer, month: string, duration: integer, campaign: integer, pdays: integer, previous: integer, poutcome: string, y: string

winequality-red.csv and winequality-red.csv: fixed acidity: double, volatile acidity: double, citric acid: double, residual sugar: double, chlorides: double, free sulfur dioxide: double, total sulfur dioxide: double, density: double, pH: double, sulphates: double, alcohol: double, quality: integer, wine_type: string

Crime_Data_from_2020_to_Present.csv: DR_NO: integer, Date Rptd: string, DATE OCC: string, TIME OCC: integer, AREA: integer, AREA NAME: string, Rpt Dist No: integer, Part 1-2: integer, Crm Cd: integer, Crm Cd Desc: string, Mocodes: string, Vict Age: integer, Vict Sex: string, Vict Descent: string, Premis Cd: integer, Premis Desc: string, Weapon Used Cd: integer, Weapon Desc: string, Status: string, Status Desc: string, Crm Cd 1: integer, Crm Cd 2: integer, Crm Cd 3: integer, Crm Cd 4: integer, LOCATION: string, Cross Street: string, LAT: double, LON: double